

ABSTRACT

CORE BANKING TRANSACTION & ACCOUNT LEDGER SYSTEM

1. OBJECTIVE

The main objective of the **Core Banking Transaction & Account Ledger System** is to develop a simplified banking application for managing customer details, bank accounts, account balances, and financial transactions in a structured manner. The system allows customer registration, account creation, deposits, withdrawals, balance enquiry, and transaction history. It demonstrates how basic banking operations can be managed using a relational database while maintaining accuracy, data consistency, and proper relationships between banking records.

2. EXISTING SYSTEM

Traditional banking systems use large and complex core banking applications to manage customers, accounts, and transactions. These systems are designed for large-scale banking operations and contain many features that are not required for a basic banking application. Managing such systems can involve complex architecture and maintenance.

The existing approach may also involve complex data management when customer, account, and transaction information needs to be connected and retrieved. Therefore, a simplified database-driven system can be used to demonstrate the basic working of core banking operations in an organized manner.

3. PROPOSED SYSTEM

The proposed **Core Banking Transaction & Account Ledger System** is developed using Python and MySQL to manage basic banking activities. The system maintains customer information, account details, account balances, and transaction records in a centralized relational database.

The major functionalities of the system include:

1. **Customer Registration** – Stores and manages customer information.
2. **Customer Management** – Allows customer details to be maintained and updated.
3. **Bank Account Creation** – Creates and links bank accounts with customers.
4. **Account Balance Management** – Maintains and updates the current account balance.
5. **Deposit Money** – Adds the deposited amount to the account balance.
6. **Withdraw Money** – Withdraws money after checking the available balance.
7. **Transaction Recording** – Records every successful deposit and withdrawal.
8. **Transaction History** – Provides details of previous account transactions.
9. **Balance Enquiry** – Displays the current account balance.
10. **Transaction Reports** – Helps retrieve and analyze transaction information.
11. **Data Validation** – Ensures valid customer, account, and transaction information.
12. **Database Relationship Management** – Maintains relationships between customers, accounts, and transactions.

4. DATABASE DESIGN

The system uses three main tables:

TABLE	PURPOSE
CUSTOMER	Stores customer information
ACCOUNT	Stores account details and balance
TRANSACTION	Stores deposit and withdrawal records

The relationship between the tables is:

CUSTOMER → ACCOUNT → TRANSACTION

A customer can have an account, and each transaction is associated with a particular account. Primary keys and foreign keys are used to maintain these relationships and ensure data consistency.

5. TRANSACTION & LEDGER MANAGEMENT

Whenever a customer performs a deposit or withdrawal, the system updates the account balance and records the transaction in the **TRANSACTION** table. The transaction record contains information such as transaction ID, account ID, transaction type, amount, and transaction date.

This transaction ledger provides a proper history of account activities and makes it easier to track financial operations.

6. DATA VALIDATION

The system validates customer and account information before performing operations. Deposit and withdrawal amounts are checked before processing. Withdrawals are permitted only when sufficient balance is available. Transactions are recorded only for valid accounts, helping maintain accurate and consistent data.

7. TECHNOLOGIES USED

- **Programming Language:** Python
- **Framework:** Flask
- **Database:** MySQL
- **Frontend:** HTML, CSS, JavaScript
- **Database Connectivity:** MySQL Connector/Python
- **Development Environment:** Visual Studio Code

8. CONCLUSION

The proposed system provides a simple and organized implementation of basic core banking operations. It manages customers, accounts, balances, and transactions through a relational database. The project demonstrates the practical use of database relationships, transaction processing, validation, and account management in a banking environment.